

1. Administrative & Study Identification

Full Study Title: Study of Pembrolizumab (MK-3475) Versus Placebo After Complete Resection of High-Risk Stage III Melanoma (MK-3475-054/1325-MG/KEYNOTE-054) **Short Title/Acronym:** KEYNOTE-054 / EORTC 1325-MG **Protocol Number:** MK-3475-054 **NCT Number:** NCT02362594 **Posting ID:** 1137004121105929006 **Trial Status:** ACTIVE_NOT_RECRUITING **Sponsor Name:** Merck (in collaboration with the European Organisation for Research and Treatment of Cancer - EORTC) **Investigational Product:** Pembrolizumab (KEYTRUDA) **Phase of Development:** Phase III (PHASE3) **Medical Monitor:** Sponsor Medical Monitor / EORTC Clinical Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether adjuvant therapy with pembrolizumab improves recurrence-free survival (RFS) at 6 months compared to placebo in the overall intention-to-treat (ITT) population of high-risk participants with stage III melanoma, as well as in the subgroup of participants with programmed cell death-ligand 1 (PD-L1)-positive tumor expression. **Secondary Objectives:** To evaluate distant metastasis-free survival (DMFS), overall survival (OS) in all patients and in the PD-L1-positive subgroup, and to assess the overall safety profile of pembrolizumab in this setting.

2.2 Background & Rationale To address the high risk of recurrence and distant metastasis in patients who have undergone complete surgical resection of high-risk stage III melanoma (stage IIIA [>1 mm lymph node metastasis], IIIB, and IIIC). The study investigates whether utilizing the anti-PD-1 therapy pembrolizumab in the adjuvant setting can prevent or delay disease recurrence and provide long-term survival benefits compared to placebo. *Note: As of Amendment 8, enrollment in Part 2 (pembrolizumab rechallenge/treatment upon recurrence) has closed, and an optional pembrolizumab extension study will not be available to participants after study closure.*

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (1:1), stratified by stage of disease and geographic region.

3.2 Planned Enrollment & Duration Target **N**: 1,019 evaluable patients **Number of Sites**: Multicenter (Global) **Estimated Study Start**: August 2015 **Estimated Primary Completion**: October 2017 (database cutoff for interim analysis) with long-term follow-up continuing for 5+ years.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years with an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 2. Complete surgical resection of Stage III cutaneous melanoma (Stage IIIA [> 1 mm lymph node metastasis], IIIB, or IIIC). 3. Tumor tissue available for evaluation of PD-L1 expression. 4. Adequate organ function. 5. No prior therapy for melanoma except surgery for primary melanoma lesions (patients previously treated with interferon for thick primary melanomas without evidence of lymph node involvement are eligible). 6. Female participants of childbearing potential and male participants must agree to use adequate methods of birth control or abstain from heterosexual activity starting with the first dose through 120 days after the last dose of study medication.

4.2 Exclusion Criteria 1. Mucosal or ocular melanoma. 2. Presence of unresectable locally advanced, in-transit, or distant metastatic melanoma. 3. History of hematologic or primary solid tumor malignancy, unless no evidence of that disease for 5 years. 4. Prior treatment with any anti-CTLA4, anti-PD-1, anti-PD-L1, or anti-PD-L2 agent, or prior participation in any Merck pembrolizumab clinical trial. 5. History of (non-infectious) pneumonitis that required steroids, current pneumonitis, or history of/current interstitial lung disease. 6. Active autoimmune disease that required systemic treatment in the past 2 years, or diagnosis of immunodeficiency. 7. Active infection requiring therapy, known history of HIV, or active Hepatitis B or C. 8. Unstable hyperthyroidism or hypothyroidism. 9. Systemic steroid therapy (or other immunosuppressive therapy) within 7 days prior to the first dose, or treatment with a live vaccine within 30 days prior to the first dose. 10. Pregnant or breastfeeding, or expecting to conceive/father children within the projected trial duration through 120 days after the last dose.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Investigational Product: Pembrolizumab **Trade Name**: Keytruda **ATC Code**: L01FF02 **Dosage/Frequency**: Pembrolizumab 200 mg or matching placebo administered once every 3 weeks (Q3W). **Route of Administration**: Intravenous (IV) infusion. **Duration of Treatment**: Up to 1 year ($\leq$ 18 doses) or until disease recurrence or unacceptable toxicity. **EMA Product Information**: [EMA Product URL](#)

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care and symptom management medications. **Prohibited:** Other concurrent investigational anti-cancer therapies, live-attenuated vaccines, and systemic corticosteroids (unless strictly used for the management of immune-related adverse events).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day -28	Informed Consent, PD-L1 status testing, Baseline imaging, Medical History
Baseline	Day 0	Randomization (1:1), First IV Infusion
Interim	Every 3 Weeks (Q3W)	IV Infusion of Pembrolizumab/Placebo, Safety Labs, irAE Assessment
End of Study	Week 52 / Recurrence	Final Treatment, Safety Follow-Up, long-term survival and DMFS tracking (up to 5+ years)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percentage of Participants With Recurrence-Free Survival (RFS) at 6 Months among all participants and among participants with PD-L1-positive tumor expression. RFS is defined as the time from randomization to the date of first melanoma recurrence (local, regional, or distant metastasis) or death from any cause, whichever occurs first. **Measurement Method:** Clinical assessment and scheduled radiographic imaging evaluated by investigators; PD-L1 expression confirmed by central laboratory. For participants who remained alive and without recurrence, RFS was censored on the date of last visit/contact with disease assessments.

7.2 Secondary & Exploratory Endpoints

- * Distant Metastasis-Free Survival (DMFS) in All Participants
- * Distant Metastasis-Free Survival (DMFS) for Participants With PD-L1-positive Tumor Expression
- * Overall Survival (OS) for All Participants
- * Safety and tolerability, with specific focus on immune-related adverse events (irAEs)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring throughout the study and for 90 days following the last dose. Specific attention is given to immune-related adverse events (irAEs) such as pneumonitis, hypothyroidism, and hyperthyroidism. **Serious Adverse Events (SAEs):** Reported to the sponsor within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** An Independent Data Monitoring Committee (IDMC) was utilized to periodically review unblinded safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: The primary efficacy analyses for RFS and DMFS were conducted on the Intent-to-Treat (ITT) population and the PD-L1 positive subgroup. **Sample Size Justification:** The study enrolled 1,019 patients to provide 92% power to detect a 10.2% increase in the 1-year RFS rate from the placebo to the pembrolizumab arm (target HR = 0.70; one-sided $\alpha = 0.014$), requiring 409 RFS events. **Handling of Missing Data:** For patients who remained alive and without disease recurrence, RFS was right-censored on the date of their last disease assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol adherence, drug accountability, and accurate AE reporting. **Audit Trail:** Validated eCRF systems with comprehensive audit trails and data clarification processes for all clinical and imaging inputs.

11. Study Identifiers & Governance

Primary ID: MK-3475-054 **Registry Number:** NCT02362594 **Sponsor/Lead Organization:** Merck and EORTC **Study Title:** KEYNOTE-054 **Official Regulatory Title:** Adjuvant Immunotherapy With Anti-PD-1 Monoclonal Antibody Pembrolizumab (MK-3475) Versus Placebo After Complete Resection of High-risk Stage III Melanoma: A Randomized, Double-Blind Phase 3 Trial of the EORTC Melanoma Group

12. Clinical Framework (Melanoma Specific)

Primary Condition: Malignant Melanoma (High-Risk Stage III) **Preferred UMLS Name:** Adjuvant Therapy for Melanoma **Condition Definition:** A malignant neoplasm derived from cells that are capable of forming melanin, which may occur in the skin of any part of the body, in the eye, or, rarely, in the mucous membranes of the genitalia, anus, oral cavity, or other sites. It occurs mostly in adults and may originate de novo or from a pigmented nevus or malignant lentigo. **Ontology Codes:** * **MeSH Code:** D008545 * **HPO Code:** HP:0002861 * **SNOMED ID:** 372244006 * **SNOMED Hierarchy:** 1240414004|Malignant neoplasm, 115223002|Nevus AND/OR melanoma, 108369006|Neoplasm **Diagnosis Threshold:** Complete surgical resection of Stage IIIA (≥ 1 mm metastasis), IIIB, and IIIC melanoma **Medical Methodology:** Post-resection adjuvant immunotherapy **Optimization Target:** Prolonged Recurrence-Free Survival (RFS) and Distant Metastasis-Free Survival (DMFS)

13. Study Procedures & Methodology

Management Pathway: Complete surgical resection followed by Q3W IV adjuvant pembrolizumab or placebo **Education Protocol (HCP):** Site initiation training focusing on the proactive identification and management of immune-related adverse events (irAEs) **Education Protocol (Patient):** Informed consent, infusion adherence, and self-monitoring for signs of immune-mediated toxicities (e.g., pneumonitis, thyroid dysfunction) **Quality Audit Mechanism:** Centralized radiographic reviews and routine IDMC safety assessments

14. Observation & Follow-Up Schedule

Total Duration: 1 year (~52 weeks) of active treatment, followed by long-term survival follow-up spanning up to 5+ years **Onsite Visit Cadence:** Every 3 weeks (Q3W) concurrent with scheduled infusions **Remote Monitoring Frequency:** Standard ongoing pharmacovigilance for survival status **Checklist Review Interval:** Scheduled radiographic tumor assessments and laboratory evaluations per protocol timelines

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				